Money can replace anything in your computer—except your beloved hard disk, which holds all your data! The venerable HDD has undergone changes over the past couple of years. Here's a shootout to help you choose your primary storage solution

Anup Nair and **Deepak Dhingra**

The Pentium I machines that hit the Indian market in the early '90s were equipped with hard disks of capacities less than 1 GB, and ran Windows 3.1. Now, a little over a decade later, it's difficult for us to find a gig of free space even on our 80 GB hard disks!

Those who've been associated with computers for the past decade would recollect the 32 GB limit—an indication of how early motherboard manufacturers underestimated the development of storage media.

We have indeed come a long way. The most recent development in the hard disk area is SATA II, a major improvement over the conventional parallel interface (IDE, short for Integrated Drive Electronics), and somewhat of an improvement over its predecessor, SATA I. Speeds have shot up from 150 MBps (in SATA I) to 300 GBps in SATA II. Technologies such as Native Command Queuing have made an entry into the SATA specification, though these require compatible motherboards for optimal performance.

Other than speeds, storage capacities, too, have been on the rise, as you know only too well. Desktop machines now feature hard disks with capacities of 250 GB and even 500 GB. You might criticise the need for such humongous capacities, but some people do need this much space, even on a desktop machine—people like gamers, professionals in animation and graphic imaging, and so on.

A PC game of today can consume two to five GB of disk space: for example, *Unreal Tournament 2004* takes up four GB. If you're using a 40 GB disk, you could complete a game, then delete the installation to install and play another one. But we don't like to do that—we like to keep everything we've got! Similarly, most of us dump movies and music onto our drives, and find it psychologically hard to delete a few to create some room.

So increased capacities and speeds it will be, and SATA is the way to go.

The test centre compared fifteen hard disks from five well-known manufacturers. Several of these were SATA II, except for Samsung's two Parallel ATA drives. SATA II drives are backward-compatible with SATA I, which is a welcome feature since some motherboards do not support the faster version. (See box *How We Tested*.)

Features

Before we settle down to comparing the drives on the basis of features, let's do away with the odd ones in the lot—the Samsung SP0411N and SV0411N. These are IDE drives with a 2 MB buffer; and moreover, the SV0411N, at 5400 rpm, was way below par. It's not really fair to compare this disk with the others, but we did it anyway—if the cost per GB turns out good, such disks can serve as large data dumps.

The contest for the top spot was very close indeed! The Hitachi drive could have been the absolute winner only if it had better shock resistance (it coul withstand 55G). The Seagate Barracuda (63G) and the Samsung SATA (63G as well) drives, too, were behind the frontrunner, WD (65G)— only in terms of shock resistance, however. The Hot-Plug feature encourages one to carry disks in and out of the machine, so high shock resistance becomes important—it is a measure of the portability of the hard disk.